Stochastic Models and Simulation

Week 1: from observed calls to a testable model

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Why this course? What do you gain?

An operations log shows what happened under one configuration. A decision often concerns a configuration never observed.

- Turn a concrete question, such as adding a bank-call agent, into outcomes and assumptions.
- Build a small Python simulation of a proposed change.
- Quantify simulation uncertainty and find assumptions that fail against real data.
- Communicate a decision, including what the available evidence cannot establish.

Your project: carry that reasoning from observed data to a defensible recommendation.

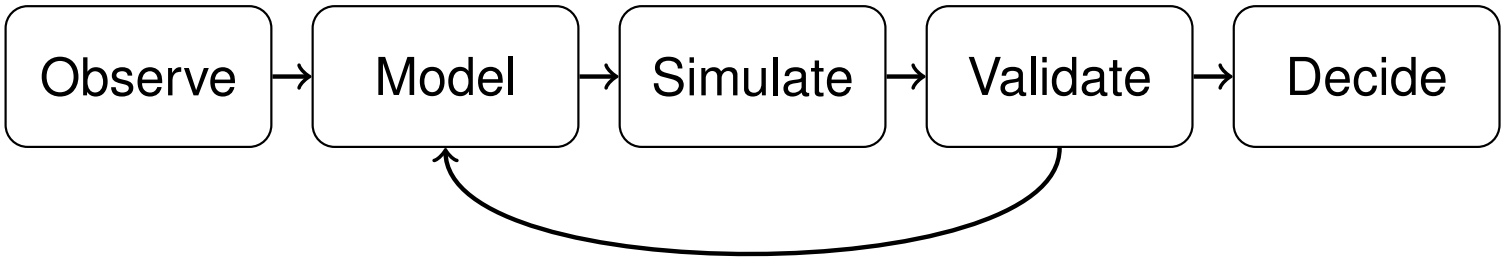
A question the existing records cannot answer

Would another agent reduce waiting at the bank's call centre?

- The records describe what happened with the staffing and routing that were in place.
- They contain no observation of the proposed staffing change.
- A simulation can explore that change, *if* its assumptions survive comparison with observed behaviour.

Today: start with arrivals. We cannot estimate the effect on waiting from arrival counts alone.

The course in one loop



- **This week:** test an arrival-count model against a real call-centre extract.
- **Across seven sessions:** add queues, estimation, uncertainty, model choice and a project decision.

Rules of the game: every three-hour session

1. **First lecture segment:** students keep devices closed. Discuss the question and work short checks on paper.
2. **Notebook and exercises:** open your own computer; use Python, inspect data and record your reasoning.
3. **Final 10 minutes in sessions 1–6:** complete a printed multiple-choice quiz individually, without devices or LLMs.

Today the opening diagnostic is part of the screen-free segment. Session 7 ends with the project, without a CC quiz.

Assessment: six paper quizzes

- One graded end-of-session MCQ in each of sessions 1–6, scored out of 20. An absence scores zero; session 7 has no CC quiz.
- Drop the *single lowest* score—your one free card. Average the other five for continuous control (CC), on the /20 scale.

$$\mathrm{CC} = \frac{\text{sum of six scores} - \text{lowest score}}{5}$$

$$\mathrm{Final} = \max\left(E,\ \frac{E + \mathrm{CC}}{2}\right)$$

Example: $0, 10, 12, 14, 16, 18 \to$ drop 0, so $\mathrm{CC} = 14$. If exam mark $E = 12$, the final mark is $\max(12, 13) = 13$.

Today's three-hour route

0–40 min	Screen-free motivation, data meaning and paper diagnostic.
40–90 min	On your computer: load, summarize and model the extract.
90–100 min	Break.
100–170 min	On your computer: simulate, validate, decide and check an LLM claim.
170–180 min	Screen-free printed quiz and one open question.

Output: a reproducible analysis and one model limitation.

Opening diagnostic: on paper, ungraded

Start with $[0, 1, 3, 1, 0, 5]$. Mark each answer *sure*, *unsure* or *guess*.

1. **Python:** CSV values are strings. How do you convert `entries` and count intervals with at least three entries?
2. **Expected versus exact:** If the model mean is two, does seeing five once disprove it?
3. **Variability:** $[2, 2, 2, 2]$ and $[0, 0, 4, 4]$ have the same mean. Which varies more?
4. **Staffing:** What evidence beyond these arrival counts would let us judge another agent?

Keep devices closed. The answers set today's pace.

The November 1999 extract: what is counted?

- File: `data/anonymousbank_nov1999_arrivals_15min.csv`
- One row per date and 15-minute interval: `date, time, entries`.
- Example: `1999-11-01,10:00,49` means 49 counted VRU entries in [10:00, 10:15).
- 2,880 intervals across 30 days; 40,722 non-`PHANTOM` *VRU entries* in total.
- The extract includes `AGENT` and `HANG` outcomes. A VRU entry is **not** necessarily a request for an agent.

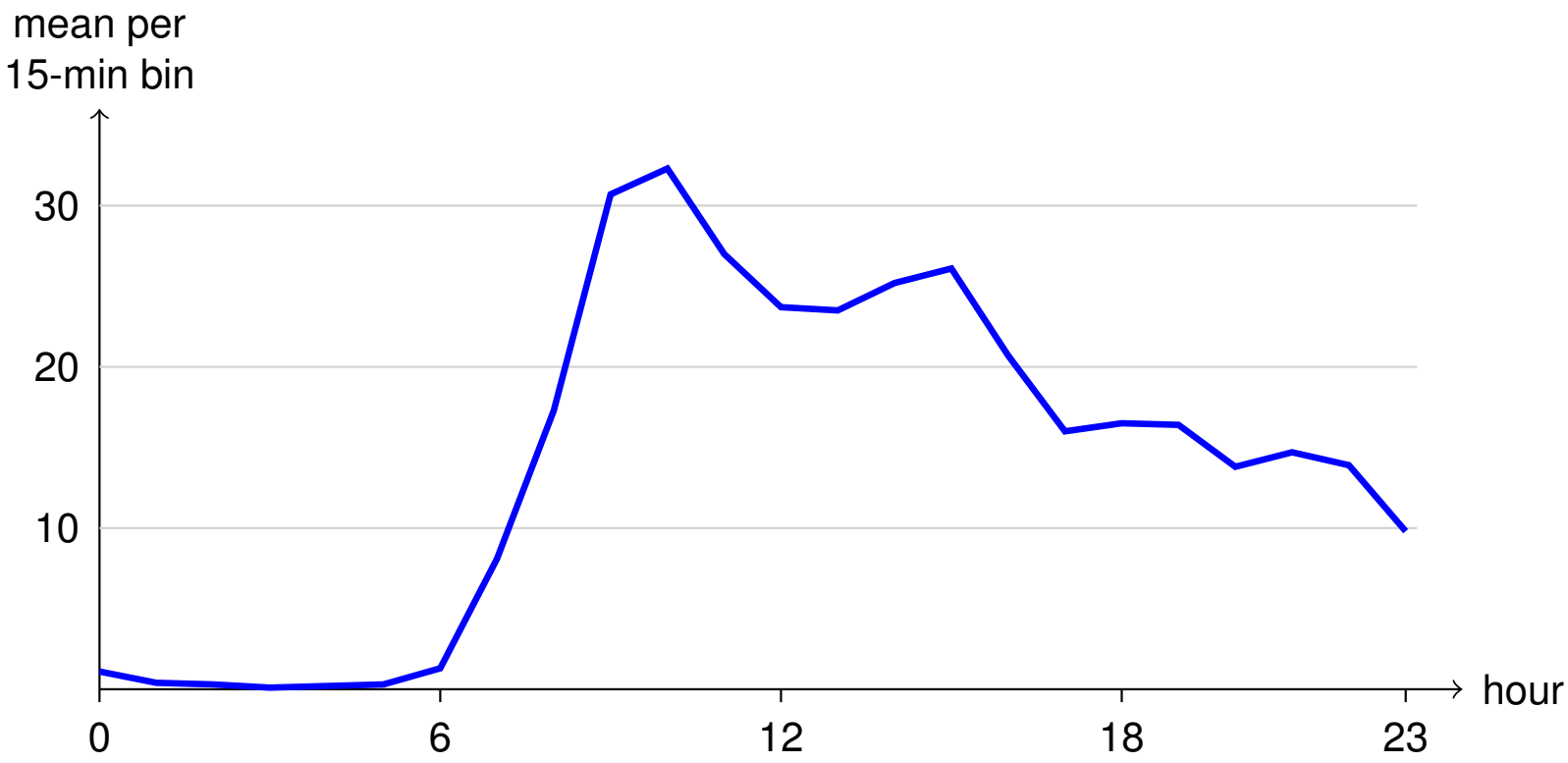
Source: Technion SEE Lab, AnonymousBank. Context: Brown et al., *JASA* 100 (2005), 36–50.

Load and check the extract

```
import csv
from pathlib import Path
filename = "anonymousbank_nov1999_arrivals_15min.csv"
path = Path("data") / filename
with path.open(newline="") as source:
    rows = list(csv.DictReader(source))
observed = [int(row["entries"]) for row in rows]
assert len(rows) == 2880 and sum(observed) == 40722
```

- Run from the repository root; the notebook also handles launch from `notebooks/`.
- Check duplicate date–time pairs, negative counts and all 96 slots per day.

Observed mean entries by clock hour



Each point averages four 15-minute slots over all 30 November days.

Discuss: Is one constant rate credible for the whole day?

First model: one arrival rate for the whole month

Assume a constant rate r VRU entries per minute. For each 15-minute count X_i:

$$X_i \sim \text{Poisson}(15r), \qquad \mathbb{E}[X_i] = \text{Var}(X_i) = 15r,$$
$$\mathbb{P}(X_i = 0) = e^{-15r}.$$

- The homogeneous Poisson process also assumes independent counts in disjoint intervals.
- Fit r from the pooled mean: $\hat{r} = 14.14/15 \approx 0.943$ entries per minute.
- Nights and busy hours are deliberately pooled. This is a baseline to test, not an endorsed staffing model.

The pooled baseline fails two checks

	Observed	**Constant-rate model**
Mean count	14.14	14.14 (fitted)
Variance of counts	209.30	14.14
Empty-bin fraction	29.1%	$e^{-14.14} \approx 7.2 \times 10^{-7}$

- The observed variance uses divisor n (Python `statistics.pvariance`).
- The pooled model predicts almost no empty bins; the extract has many.
- This challenges a *constant* rate. We must inspect when the empty bins occur.

Simulate arrivals with plain Python

```
import random, statistics
rate = statistics.mean(observed) / 15  # per minute
rng = random.Random(2026)
simulated = [0] * len(observed)
clock = rng.expovariate(rate)
while clock < 15 * len(observed):
    simulated[int(clock // 15)] += 1
    clock += rng.expovariate(rate)
```

Each exponential gap moves the clock to the next arrival. The notebook wraps this in a reusable function with input checks.

Check the simulator; then check the model

- With a few different seeds, simulated mean counts should stay near 14.14.
- The simulated variance/mean ratio should be near 1, not exactly 1 in every run.
- Simulated empty-bin share should be near $e^{-14.14}$; the observed share is 29.1%.

Do not tune the seed to make the simulation resemble the data. This is a test of the constant-rate assumption, not a search for a lucky run.

Validate by clock hour, then propose a next test

- The homogeneous model expects about 14.14 entries per 15-minute bin at *every* clock hour.
- Observed means: about 0.1 at 03:00 and 32.3 at 10:00, across all November days.
- Find where the notebook's simulated hourly profile overpredicts and underpredicts the data.

Next test: estimate separate rates for comparable time slots and day types; reserve later days to check them. A better hourly profile alone would not prove independent Poisson arrivals.

What can we decide today?

Supported now

- Describe when VRU entries occur.
- Test a simple model for entry counts.
- Decide which assumption needs a better check.

Needed for staffing

- Agent-directed arrivals and routing.
- Service durations and staffing schedules.
- Waiting, abandonment and model validation.

Decision for week 1: identify the missing observations before judging whether another agent would help.

Use an LLM to check a theoretical claim

Ask an LLM:

> *"If counts in periods A and B are each Poisson, is a count from a randomly selected period Poisson with their average rate? Prove or disprove it. Distinguish selection from addition."*

1. Use a fair coin to select A or B, with $A \sim \text{Poi}(1)$ and $B \sim \text{Poi}(3)$.
2. Check the selected count's mean and variance yourself. Compare with $\text{Poi}(2)$.
3. State separately what happens if independent counts A and B are *added*.

Record the LLM claim, its assumptions, your independent check and any correction. Use no raw call records in the prompt.

End-of-session participation check

In the final 10 minutes, complete the printed six-question exit quiz handed out in class.

- Work individually, without devices or LLMs.
- Choose one answer per question.
- Give one question about the data or model that you still have.
- Hand it in before the group discussion.

The quiz checks measurement, a basic Python conversion, variability and independent LLM verification.

What we can take forward

- The bank extract gives a real arrival pattern, with important limits on what it measures.
- A simple Poisson model is a useful starting point only when its rate and independence assumptions are stated and checked.
- Simulation creates a testable prediction; observed counts show where the homogeneous baseline fails.
- A staffing recommendation needs queue, service and abandonment evidence too.

Sources: Technion SEE Lab, AnonymousBank data repository; Brown et al., “Statistical Analysis of a Telephone Call Center,” *JASA* 100 (2005), 36–50.